HF SIGNAL 01: A Candidate Machine-Readable Protective Signal for Testing and Standardisation

Technical Parameters, Legal Limits and Validation Requirements

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Abstract

The prohibition on attacking a person who is recognised as hors de combat is a rule of customary international humanitarian law (Rule 47 of the ICRC Customary IHL Study; Article 23(c) of the Hague Regulations of 1907), binding on all parties to an armed conflict irrespective of treaty ratification. For States party to Additional Protocol I, Article 41 provides its clearest treaty formulation, while Article 57 requires those who plan or decide upon an attack to do everything feasible to verify that the attack remains lawful. While these obligations have long been in force, autonomous and semi-autonomous systems operating at machine speed create a practical problem: no commonly specified machine-readable form exists through which an intention to surrender can be communicated to, and recognised by, such systems.

This paper presents HF SIGNAL 01, a candidate surrender-recognition protocol intended as a technical referent for machine-readable expressions of surrender. The protocol offers two alternative modalities: a passive visual modality, requiring no equipment, and an active electronic modality comprising two channels that must be confirmed together, a pulsating white optical signal and a radio-frequency signature. The paper does not claim operational effectiveness, adoption, or legal status. The protocol is a supplementary means of expressing an intention to surrender, never a condition of protection, and its absence permits no adverse inference. Instead, the paper evaluates the proposal against previously defined feasibility criteria for lawful operational safeguards, including technology readiness, latency, annotated datasets, and independent validation.

The assessment is deliberately conservative. HF SIGNAL 01 currently exists as a fully specified and publicly documented concept without experimental validation, implementation, or performance data. Its present contribution is documentary rather than operational. The paper argues that the existence of a publicly available and integrity-protected candidate standard changes the factual context in which surrender-recognition capabilities may be assessed, transforming what was previously described solely as a technical impossibility into a documented question of design choice, review, and validation. The paper concludes by inviting independent technical testing, dataset development, and examination within relevant international humanitarian law and autonomous weapons governance fora.

Keywords: international humanitarian law; autonomous weapons systems; surrender recognition; hors de combat; Article 41 Additional Protocol I; lawful operational safeguards; machine-readable signals; CCW GGE.

1. The Legal Framework and the Effectiveness Gap

An earlier paper by this author defined a lawful operational safeguard as a technical measure that implements, within the operational architecture of an AI system, an obligation already binding under international humanitarian law — as distinct from safeguards that reflect discretionary ethical preference, reputational concern or policy choice.[1] This paper does not restate that framework. It applies it to a single artefact, and it therefore begins with only the anchoring provisions.[2]

The underlying obligation to protect persons hors de combat is customary (ICRC Rule 47; Hague Regulations 1907, Art. 23(c)), binding on all parties to an armed conflict. Its clearest treaty formulation, which this paper uses to anchor the technical analysis of machine recognition, is found in Additional Protocol I:

> 1. A person who is recognised or who, in the circumstances, should be recognised to be hors de combat shall not be made the object of attack.
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> 2. A person is hors de combat if: (a) he is in the power of an adverse Party; (b) he clearly expresses an intention to surrender; or (c) he has been rendered unconscious or is otherwise incapacitated by wounds or sickness, and therefore is incapable of defending himself; provided that in any of these cases he abstains from any hostile act and does not attempt to escape.[3]

Article 57(2)(a)(i) requires that those who plan or decide upon an attack shall "do everything feasible to verify that the objectives to be attacked are neither civilians nor civilian objects[4]...".

Three features of these provisions carry the whole argument. First, they are in force: both have been treaty law since 1977 (in force from 1978), and customary law long before, and neither depends for its validity on any development in weapons technology. That this remains so for systems of the kind considered here is not contested in the forum where they are discussed: the Guiding Principles affirmed by the Group of Governmental Experts state that international humanitarian law continues to apply fully to all weapons systems, including the potential development and use of lethal autonomous weapons systems. The principle settles the applicability of the law; it says nothing about the questions this paper raises, which begin after it.[5] Second, they are means-neutral: they attach to the party conducting the attack, whatever instrument it uses. Third, and decisively for present purposes, Article 41(2)(b) presupposes an act of communication. A person is hors de combat under that limb because he "clearly expresses an intention to surrender"; the provision protects an expression, and an expression succeeds only if its addressee can read it. The 1987 ICRC Commentary already recognised that the form of this expression is not fixed: a formal surrender is not always realistically possible, and the intention must be conveyed by whatever means the circumstances allow.[6]

This is where the gap opens. When the addressee of the expression is a human being, a raised white cloth, raised hands or a discarded weapon can be read within the time a human decision takes. When the addressee is a system that detects, classifies and engages within milliseconds, there is no guarantee that any human-legible gesture is read at all before the engagement decision is executed — and there is currently no shared, machine-readable code through which an expression of surrender could be conveyed and read at that speed. The flexibility the Commentary allows is itself conditioned on the addressee. A human reader can interpret without any predefined code: a raised white cloth means what it means because a person supplies the meaning. A machine perceives only what has been defined for it in advance, and an expression for which no class has been defined is not misread — it is not read at all. What is missing at machine speed is therefore not only time but form: a fixed, published referent that perception can be built to recognise. The difficulty is not asserted here for the first time: the obstacles to reliable recognition of surrender by autonomous systems — the variability of its expression,

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[1] Giovanni Nardacci, Lawful Operational Safeguards in AI Systems: Surrender Recognition, Compliance Architecture, and International Humanitarian Law (Fifth Edition, 2026), Zenodo DOI 10.5281/zenodo.21932439; SSRN Abstract ID 6821838 [hereinafter "Paper I"]. The canonical definition of "lawful operational safeguard" is given in Section VI.A; the feasibility criteria in Section V.A; the treatment of failure modes in Section VII.

[2] Protocol Additional to the Geneva Conventions of 12 August 1949, and relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977, 1125 UNTS 3 [hereinafter "AP I"]. All treaty provisions quoted in this paper were verified verbatim against the official text as published in the ICRC IHL Treaties Database, ihl-databases.icrc.org.

[3] AP I, Art. 41(1)–(2).

[4] AP I, Art. 57(2)(a)(i).

[5] Guiding Principles affirmed by the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems, Annex IV, CCW/GGE.1/2019/3 (2019), Principle (a): international humanitarian law continues to apply fully to all weapons systems, including the potential development and use of lethal autonomous weapons systems.

[6] Yves Sandoz, Christophe Swinarski and Bruno Zimmermann (eds.), Commentary on the Additional Protocols of 8 June 1977 to the Geneva Conventions of 12 August 1949 (ICRC/Martinus Nijhoff, 1987), para. 1612 (on Art. 41: a formal surrender is not always realistically possible; the expression of the intention to surrender depends on the means available in the circumstances).

its dependence on context, and its performance under adversarial conditions — are documented in the independent literature.[7] The result is not a normative gap. The norm is intact. What is missing is a means of giving effect to it. The same structure — an existing obligation resting on an implicit assumption about human reaction time, which a class of machines silently removes — was analysed for civilian machine-safety law in the author's second paper.[8]

What follows from this is not a new rule, and it is not — yet — a standard. It is an enumeration of principal response pathways. The obligation is unconditional; the available paths by which a party deploying machine-speed systems can perform it include: (1) restoring meaningful human decision time through genuine supervision or doctrinal restriction of autonomy; (2) training machine perception to recognise existing human expressions of surrender through validated behavioural recognition; (3) introducing a shared machine-readable referent — a public, dated, technically specified code that machine perception can be built to read; or (4) restricting the system's targets, environments, speed or modes of operation so that legally protected changes of status remain capable of affecting conduct before irreversible force. These are the principal pathways, and they may be combined; if another exists that produces the same effect, it stands on the same footing. A party that follows none of them is not choosing an alternative; it is not performing the obligation. They are not, however, equally available today, and the difference is one of state of the art rather than of principle. The first turns on doctrinal and command decisions rather than on any technical development, and where it is taken it dissolves the problem this paper addresses. The second is legitimate in principle, and nothing here forecloses it, but it remains an open research programme rather than an available means: no generally accepted specification of the behaviours to be recognised, and no validated general-purpose recogniser, has been identified in the literature reviewed here, and the difficulties — the variability of the human repertoire across cultures and circumstances, and its performance under adversarial conditions — are documented in the independent literature. The fourth restricts the capability itself, which is a legitimate answer but one that may entail significant operational cost and may therefore be difficult to sustain under operational pressure. For present purposes, the third pathway has three properties that make it particularly amenable to ex ante specification, testing and legal review; the case for it rests on these rather than on any merit of a particular specification. First, it does not require contextual interpretation of an open-ended human behavioural repertoire. The second asks machine perception to read a human repertoire that is variable and context-dependent; the first and the fourth depend on judgements made elsewhere about whether supervision is real or restriction adequate. A fixed referent asks only whether a defined signature is present, which is a question of measurement rather than of understanding. Second, and in consequence, it is particularly amenable to direct, quantitative verification before the encounter: whether a given system detects a defined signature, at what range, with what latency and with what error rate, can be established in advance, whereas supervision, behavioural recognition and operational restriction are less reducible to a bounded pre-encounter test and remain more dependent on context, generalisation and operating conditions. Third, it need not require the prior removal of autonomy or a categorical narrowing of the system's intended employment: it adds a detector rather than subtracting a capability, and can therefore be adopted without a prior policy decision, though not without operational cost. A corollary follows for legal review. Because a referent can be specified in advance, its absence from a system can also be established in advance — which is what makes the question examinable at procurement rather than reconstructible only after an engagement. A further consequence bears on the systems at issue here. The first pathway presupposes a human in the engagement loop, whose decision

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[7] The difficulty is independently documented. See Robert Sparrow, "Twenty Seconds to Comply: Autonomous Weapon Systems and the Recognition of Surrender", 91 International Law Studies 699–728 (2015) (U.S. Naval War College), analysing why reliable recognition of surrender by autonomous weapon systems remains unresolved and why systems lacking it are likely to remain controversial; and ICRC, ICRC Position on Autonomous Weapon Systems (12 May 2021), with background paper, observing that employing anti-personnel autonomous weapon systems that are not capable of recognising an enemy's clear expression of an intent to surrender, and of refraining from engagement, raises serious IHL concerns — and recommending constraints on such systems accordingly. On the content and practical conditions of the rule of surrender itself, see Russell Buchan, "The Rule of Surrender in International Humanitarian Law", (2018) 51(1) Israel Law Review 3–27; on the IHL compliance questions raised by autonomous weapon systems generally, see Laura Bruun, Marta Bo and Netta Goussac, Compliance with International Humanitarian Law in the Development and Use of Autonomous Weapon Systems: What does IHL Permit, Prohibit and Require? (SIPRI, March 2023).

[8] Giovanni Nardacci, Systemic Arrestability: Operator Protection in Machine Safety Law When Machine Speed Exceeds Human Reaction, Zenodo, DOI 10.5281/zenodo.20837150; SSRN Abstract ID 6923461 [hereinafter "Paper II"]. Paper II applies the same method to civilian machine-safety law: an existing obligation rests on an implicit temporal assumption that a class of machines eliminates.

time is restored; the fourth narrows the system's permitted employment. Where a system is designed to operate without such a human, and where its intended employment is not to be narrowed, the capacity to register a protected condition has nowhere left to reside except within the system itself. That leaves the two system-internal pathways — behavioural recognition and a fixed machine-readable referent — as the relevant candidates, and between them the difference is the one identified above: the detection performance of a fixed referent can be specified and quantitatively tested in advance against a defined signal, whereas the generalisation of a behavioural recogniser cannot be established in the same bounded way.

HF SIGNAL 01, described in the next Section, is a candidate for the third pathway. It is not presented as the solution, nor as superior to the other paths. It is the first concrete attempt known to the author to specify a machine-readable referent in full technical detail, and it is offered here for scrutiny — including hostile scrutiny — rather than for adoption.

2. The Protocol: HF SIGNAL 01 Technical Specification

2.1 Dual-channel architecture

HF SIGNAL 01 is a dual-channel signal. The full specification is the protocol issued on 24 April 2025 and preserved in an integrity-protected file; it is summarised in the Technical Note of May 2026,[9] which is the Association's current statement of the specification.[10] Recognition is valid only when both channels are confirmed:

Optical channel. A pulsating white light (colour code #FFFFFF) at a pulse frequency of 7 Hz, uniform pulse pattern, with a recommended brightness of at least 300 lux at the source, a nominal visibility range of 100 m, and infrared compatibility. A 7 Hz periodic signature is deliberately simple: it sits well below the Nyquist limit of ordinary camera frame rates, so that no specialised optics are required to detect it — only software configured to look for it.

Radio channel. A continuous FM tone at 144.0 MHz — within the amateur VHF band — with a bandwidth of 25 kHz and a tone frequency of 1,000 Hz; the specification indicates a detection threshold of −85 dBm and recommends threshold crossing as the detection method.

The redundancy rule is explicit in the specification ("dual_channel_required"): a system must confirm both the optical and the radio signature before treating the signal as present. The conjunction is the protocol's principal defence against accidental activation. Emission occurs in ten-second cycles and may be activated manually or automatically.

Design rationale. The choices embodied in the specification follow from constraints implicit in its purpose. The controlling constraint is on the recognising side: the signal must be readable by sensor classes already ubiquitous on autonomous platforms — cameras and general-purpose radio receivers. This excludes QR-type visual codes, RFID, Bluetooth, and encrypted transponders. The second constraint is on the emitting side — minimal, cheap, powered hardware — and the third is redundancy across two independent physical channels. Within these constraints, the specific numerical values are set by engineering judgment. Their merit at this stage is that they are fixed and therefore testable.

Earlier design iterations of the signal exist in the public history of the Association's repositories. However, as clarified in the accompanying Architecture Note, the protocol comprises two alternative modalities: the active electronic dual-channel specification described in this section, and a passive visual modality (a white disc on a red field). The passive visual modality is a current element of the protocol, not a superseded design stage; it is the modality available to a person without powered equipment. The two modalities are alternative expressions of the same protective meaning.

[9] Human Flag Association, HF SIGNAL 01 — Technical Note on AI Safeguards and International Humanitarian Law (May 2026), HF_SIGNAL_01_Technical_Note_May2026.md, in the public repository github.com/nardaxxx/hf-signal-01. The Technical Note summarises the specification; the full parameter set is in the integrity-protected file cited next.

[10] Human Flag Association, HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt (issued 24 April 2025), in the repository cited supra note 9; integrity protected by SHA-256 hash 3a7de50d3cec9843aae11fcfe1c295f7ffb8e7eac3cbfc5851a43b509812182e and OpenTimestamps certification (HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt.ots). All parameter values quoted in this Section are taken from that file and from the Technical Note, supra note 9.

2.2 Operation in practice

The specification does not stop at the physical signature; it defines when the signal is emitted, by whom, and what a recognising system is expected to do. These operational provisions matter for the legal analysis, because they show the protocol as a complete act of communication rather than a bare waveform.

Activation. Three activation modes are defined. The first is manual: the person facing a perceived imminent threat activates the signal deliberately, with a required confirmation step so that emission is a conscious act. The second is proximity-triggered: activation on detection of an autonomous system or unidentified threat within a nominal 50-metre radius, using infrared, visual, motion or RFID sensing on the emitter side. The third is preprogrammed periodic broadcast — at a nominal interval of fifteen minutes — during exposure to a known high-risk environment, for instance movement through an area where autonomous or semi-autonomous systems are known to operate. The specification names as intended emitters a human operator, an unarmed drone and a civilian device, and as intended environments the combat zone, the urban conflict area and the automated surveillance area. It also states a condition that deserves emphasis: activation must occur in good faith and under perceived threat. The signal is an expression of protected status, not a general-purpose beacon, and the specification treats it as such.

Expected response. On the recognition side, the specification defines the expected behaviour of a system that has confirmed both channels: immediate cessation of aggressive behaviour, initiation of a non-hostile identification procedure, and prioritisation of the preservation of human life. Two qualifications are required, and the reader should hold both. First, these are specification-side expectations — statements of what compliant recognition logic is expected to do — not descriptions of any implemented behaviour; no system implementing them exists (Section 3). Second, they map directly onto the structure of Article 41: cessation of engagement gives operational effect to the words "shall not be made the object of attack", and a subsequent identification procedure carries out the verification that Article 57(2)(a)(i) requires. The expected-response clause is, in other words, the point at which the technical document and the treaty text meet.

Emitter hardware. The emitter side is deliberately low-technology: a white LED or xenon source capable of a 7 Hz pulse, and a low-power VHF transmitter capable of a continuous 1,000 Hz FM tone at 144.0 MHz. Nothing in the specification requires bespoke equipment, and this is a design choice — a standard that only well-resourced actors could emit would fail its own purpose. It remains true, however, that the protocol presupposes possession of a functioning, powered emitter, and that presupposition is itself a limit of the protocol; it is treated as such in Section 5.

2.2a Legal classification of activation modes

The legal significance of the signal depends on the mode of activation. Manual activation by a person, or by a device acting on that person's deliberate and contemporaneous instruction, may serve as evidence of an expressed intention to surrender within the meaning of Article 41(2)(b) of Additional Protocol I, provided that the surrounding circumstances confirm the expression is genuine and that the person abstains from hostile acts.

Automatic, proximity-triggered or periodic activation does not by itself establish such an intention and is not characterised by this specification as an act of surrender. It functions only as a precautionary protective alert requiring temporary suspension of engagement, human or system-level verification, and assessment of the surrounding circumstances. Signals originating from unmanned devices likewise do not confer hors de combat status upon an object; they may only communicate on behalf of identifiable persons or trigger a precautionary response.

This distinction is not a defect of the specification but a necessary consequence of the underlying law: a machine-readable signal can express an intention only when a human intention is in fact being expressed. Where it is not, the signal retains its precautionary protective function but does not engage the specific protections attaching to surrender or hors de combat status.

2.3 Direction of the signal: emitter and addressee

The structure of surrender involves two distinct roles that are easily conflated: the one who surrenders, and the one who must recognise the surrender. Article 41 places the legal burden on the second role — it is the attacker who must not attack a person who "should be recognised" to be hors de combat. HF SIGNAL 01 addresses the machine instantiation of that second role by giving the first role a code the second can read: the signal is emitted by the protected person in order to be recognised by the system. The direction runs from the protected person to the machine. It is a signal of recognition, not an instrument of targeting.

Two clarifications follow. First, the signal confers no status by itself. A person is hors de combat because the conditions of Article 41(2) are met, however expressed; HF SIGNAL 01 adds a machine-legible means of expression and is not a source of protection independent of the norm. The design rule that operationalises this is stated once here and holds throughout: compliant recognition logic treats a confirmed signal as a sufficient condition for an immediate protective response — suspension or inhibition of force, preservation of the signal and relevant sensor data, and verification of the surrounding circumstances — but not, by itself, as conclusive proof of hors de combat status. The signal is never a necessary condition of protection: its absence permits no adverse inference whatsoever, and the full repertoire of human-legible expressions of surrender retains its entire legal force. Any implementation that inverted this logic — treating non-emission as an indicator of targetability — would violate Article 41, not implement it. Second, HF SIGNAL 01 is not a recognised emblem, sign or signal within the meaning of Article 38 of Additional Protocol I,[11] and this paper claims no such status for it. It is a candidate technical standard, whose legal significance is entirely derivative of the existing obligation it would render performable.

2.4 The ethical clause of the 2025 file: scope and legal nature

The integrity-protected specification contains an ethical clause that a careful reader will find and should find. Its "forbidden_use" list includes "military targeting systems" and "weapons", and its declaration states that use of the signal "for military targeting, suppression, deception, or monetisation constitutes a violation of international humanitarian ethics[12]". Read in isolation, this could be taken to prohibit precisely what this paper appears to discuss — the processing of the signal by weapon systems. Two things resolve the apparent tension: the direction of use, and the legal nature of such clauses.

The prohibition targets offensive instrumentalisation of the signal: feigning it in order to invite reliance and then strike; exploiting its emission to locate, identify and engage the emitter; feeding it into an engagement chain as targeting input. In each of these cases the signal is turned against the person it exists to protect. What this paper discusses is the opposite direction: that a system recognise the signal in order to abstain. Recognition-for-abstention does not "use the signal for targeting"; it makes the signal legible, which is its function — in the same way that teaching a soldier to recognise the white flag is not a military use of the white flag. The distinction tracks the two roles identified in Section 2.3: the prohibition protects the one who surrenders; recognition capability is demanded of the one who must recognise.

As to its legal nature, the clause belongs to the April 2025 file, which is integrity-protected and therefore cannot be amended, and it reflects the protective intent at the moment of issuance. The Technical Note of May 2026, which summarises the specification, does not restate it. This is not an oversight; it is a clarification of what such a clause can and cannot do. For a signal, a prohibition written by its issuer is declarative, not constitutive: the white flag carries no licence terms, and its protection against abuse derives from the law itself — the prohibition of perfidy and of the improper use of signs and signals — which operates, as Section 5 shows, irrespective of any clause the issuer might attach and irrespective of the signal's status. The 2025 clause therefore adds emphasis, not obligation, and nothing in the analysis of this paper depends on it. For the record, its reading is nonetheless stated plainly: implementing recognition of HF SIGNAL 01 for the purpose of withholding or breaking off engagement is the intended use of the specification; any processing of the signal that contributes to the selection or engagement of the emitter as a target is abuse — condemned by the clause and, where it

[11] AP I, Art. 38(1) (prohibition of improper use of recognised emblems, signs and signals).
[12] HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt, supra note 10, ethical_clause.

meets the elements of Article 37, prohibited by law. Separately, and on a different plane altogether, the specification document is distributed under CC BY-NC-ND 4.0, free for humanitarian and civilian use, with institutional use subject to a written licence[13]; this is a sustainability model for the issuing association and has no bearing on the analysis, under international humanitarian law, of recognition of the signal.

2.5 Precedent within the treaty architecture: distinctive signals

A reader encountering a light-plus-radio protection signal for the first time might take the concept itself to be novel. It is not, and the treaty says so. Additional Protocol I defines, in Article 8(m), the "distinctive signal" — "any signal or message specified for the identification exclusively of medical units or transports" in Chapter III of Annex I to the Protocol — and Article 18(5)–(6) authorises Parties to use such signals to identify medical units and transports. Annex I, in its amended form, specifies those signals technically: a flashing blue light with a recommended flash rate of between sixty and one hundred flashes per minute, a radiocommunication signal, and electronic means of identification.[14] In other words, the architecture of the Protocol already contains protection-relevant communication in the form of technically specified luminous and radio signatures, designed to be recognised by instruments as well as by human observers, with parameters fixed in a technical annex and maintained over time — Annex I was revised in 1993 precisely to keep the specifications current.

The comparison must be drawn with care, because it proves something narrow. HF SIGNAL 01 is not a distinctive signal within Annex I, has no status under it, and borrows none of its protection — consistently with the position already taken on Article 38 in Section 2.3. What the comparison establishes is only this: the kind of object HF SIGNAL 01 proposes to be is a kind the Protocol already knows. A specified pulse frequency for a light and a specified carrier for a radio emission, serving the practical operability of a protective rule, is not an alien graft onto international humanitarian law; it is the same technique the Conventions' own machinery uses for the medical function, applied here to the expression of surrender at machine speed. Everything the distinctive signals have and HF SIGNAL 01 lacks — State negotiation, treaty authorisation, institutional maintenance — is not a defect of the comparison but a description of the distance still to be travelled: it is precisely the gap that the invitation of Section 6 asks the competent institutions to close. And the Annex I precedent shows the institutional path such specifications travel — proposed, technically reviewed, amended, maintained by competent bodies. That is the path that Section 6 proposes for this candidate.

3. Feasibility Mapping against the Criteria of Paper I, Section V.A

Paper I sets out feasibility criteria that any candidate lawful operational safeguard must eventually satisfy: technology readiness, latency thresholds, annotated datasets and independent validation.[15] A paper that presents a candidate and then exempts it from the author's own criteria would not deserve to be read. This Section therefore assesses HF SIGNAL 01 against each criterion without modification. The summary is in the table; the paragraphs that follow give the reasons.

Criterion (Paper I, §V.A)	What the criterion requires	Status of HF SIGNAL 01 (July 2026)
Technology readiness	Demonstrated maturity of the recognition capability, at minimum analytical or experimental proof of concept.	TRL 2. Concept and full parametric specification exist; no proof of concept, no experiment, no measurement.
Latency thresholds	Measured detection latency compatible with the engagement timeline of the systems concerned.	No latency measurement exists. None is claimed.

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[14] AP I, Arts. 8(m) and 18(5)–(6); Annex I to AP I, Regulations concerning identification (as amended on 30 November 1993), Chapter III (light signal, radio signal, electronic identification).
[15] Paper I, supra note 1, Section V.A.

Criterion (Paper I, §V.A)	What the criterion requires	Status of HF SIGNAL 01 (July 2026)
Annotated datasets	Curated positive and negative samples across operational conditions, sufficient to train and evaluate detectors.	None exist. The specification's reference to a YOLOv8-class model is a recommendation; no model has been trained.
Independent validation	Testing by a party other than the proponent — laboratory, standards body or equivalent.	None has been performed. Deposit and timestamp establish provenance, not validity (Section 4).

Technology readiness. HF SIGNAL 01 stands at TRL 2: a technology concept formulated.[16] No prototype has been built to the specification, no field or laboratory trial has been run, and no performance figure of any kind has been produced. TRL 3 requires analytical or experimental proof of concept; neither exists, and this paper claims neither. Within that level, one property is worth stating separately: the specification is parametrically complete. Pulse frequency, colour code, brightness, visibility range, carrier frequency, bandwidth, tone frequency, detection threshold and the redundancy rule are all numerically fixed, which makes the protocol directly testable by any third party without further design work. Completeness of specification is a genuine strength at this stage — and it is not evidence of performance.

Latency. The criterion asks whether detection can occur within the decision timeline of the systems concerned. For HF SIGNAL 01 no latency has been measured. A plausibility argument can be made — a 7 Hz signature is in principle confirmable within a few pulse periods, that is, within several hundred milliseconds of observation, and threshold detection of a continuous tone at −85 dBm is routine practice in software-defined radio — but plausibility is not measurement, and the difference between the two is exactly what independent testing exists to establish. This paper offers no latency claim.

Annotated datasets. No dataset exists: no corpus of positive and negative optical samples across distance, weather, illumination, motion and occlusion; no library of radio captures across interference and multipath conditions; no adversarial samples. Consequently no detector has been trained, and the false-positive and false-negative rates of the protocol — the numbers on which any deployment argument would ultimately turn — are simply unknown. Dataset construction is the single most consequential open task, because every other criterion depends on it.

Independent validation. None has taken place. No university laboratory, standards organisation or other third party has examined the protocol. The documentary apparatus described in Section 4 — deposit, hash, timestamp — establishes that the specification has existed, in exactly this form, as of a certified date. It establishes nothing about whether the specification works. The two kinds of evidence must not be confused, and this paper does not confuse them.

Against its own framework, then, HF SIGNAL 01 currently satisfies only the precondition that the framework's criteria presuppose: the existence of a fixed, public candidate that is specified closely enough to be tested. The qualification matters and is stated here rather than left to the reader: the active electronic modality is parametrically complete and directly testable, while the passive visual modality is architecturally defined but awaits normative parameterisation of the values listed in Section 5. Everything else remains to be done. So that "everything else" is not left vague, the validation task can itself be specified.

What validation would require. For the optical channel: a factorial capture campaign producing positive samples across a range of distances (up to and beyond the nominal 100 m), illumination regimes (daylight, low light, night, and infrared), weather conditions, emitter motion, and partial occlusion; a negative corpus of confusable periodic light sources — emergency strobes, damaged luminaires, screen flicker, and reflections — recorded under the same conditions; and an adversarial set of deliberate imitations. For the radio channel: recordings of the specified tone across urban multipath, co-channel amateur traffic, interference and jamming conditions, together with negative captures of ordinary 144 MHz activity. On this material, a reference detector — built from off-the-shelf components, a standard camera and a software-defined radio, precisely because the specification demands nothing more —

[16] The technology readiness level (TRL) scale is used here in the sense of ISO 16290:2013, Space systems — Definition of the Technology Readiness Levels (TRLs) and their criteria of assessment, whose definitions are in general engineering use. TRL 2 corresponds to a technology concept formulated; TRL 3 requires analytical or experimental proof of concept.

would yield the three numbers on which everything turns: a false-positive rate, a false-negative rate, and a time-to-confirmation distribution measured from signal onset, per channel and for the required conjunction of both. The campaign must be preceded by an explicit adversarial model, stating the assumed adversary capabilities — jamming of either channel, replay, synthesis of the radio signature, optical imitation, coordinated decoys — because false-positive and false-negative rates are meaningful only relative to a stated threat, and a validation that omitted the adversary would validate nothing. Replication of those numbers by a party other than the proponent would complete the step from TRL 2 to demonstrated proof of concept. None of this requires the cooperation of any weapons manufacturer or any State: it is ordinary machine-perception research that can be carried out by a single university laboratory, and it is the concrete content of the invitation extended in Section 6.

One further open issue belongs to any future standardisation process rather than to this paper: the protocol's radio channel sits in an allocated amateur band, and the choice deserves to be critically examined. Spectrum regulation differs across jurisdictions; the band carries legitimate traffic and may be congested; and in a conflict zone it may be deliberately jammed or unavailable, while emission by unlicensed persons raises regulatory questions of its own that would require coordination with the competent spectrum authorities. The designated carrier raises two further difficulties of different kinds. The first is regulatory: 144.0 MHz is the lower edge of the amateur allocation, so an emission 25 kHz wide and centred on it places roughly half its bandwidth below 144 MHz, in spectrum allocated not to the amateur service but to other services, which differ by ITU Region. The second concerns coordination rather than law: the opening segment of the 2-metre band is reserved by IARU regional band plans for narrow-band modes — in Region 1, 144.000–144.135 MHz is reserved exclusively for telegraphy — so a wideband FM emission would not conform to those plans even if it remained within the allocation; the segments allocated to FM telephony lie in the upper part of the band. Neither difficulty affects the detectability of the signal, and neither is answered by this paper; both indicate that the carrier is a value to be fixed by coordination, and that a carrier placed clear of the band edge would remove the first difficulty without altering anything else in the specification. A standardisation body would also be the proper forum for deciding whether 144.0 MHz should remain the designated carrier or whether a dedicated protected allocation — the treatment that Annex I signals received for the medical function — is warranted; the specification fixes a testable value, it does not foreclose that discussion.

4. Provenance and Documentary Integrity

The chronology of the specification is short and fully documented. The full protocol was completed on 24 April 2025. It was deposited with the Legal Services of the State Council of the Canton of Ticino on 28 April 2025[17] and published in May 2025 in a public repository. The protocol file is protected by the SHA-256 digest 3a7de50d3cec9843aae11fcfe1c295f7ffb8e7eac3cbfc5851a43b509812182e and by an OpenTimestamps certification anchoring that digest in the Bitcoin blockchain,[18] so that any reader can verify, without trusting the issuing association, that the file examined today is byte-for-byte the file that existed at the certified date. The Technical Note of May 2026 summarises the specification; the parameter values are those of the integrity-protected file.

The function of this apparatus must be stated precisely, because it is easily overstated. Deposit, hash and timestamp establish two things: intellectual priority — the specification, with these exact parameters, existed publicly from these dates — and integrity against subsequent alteration — it cannot have been silently altered since. They establish nothing else. In particular, as Section 3 has already said, they are not evidence of technical validity, and they are not presented as such.

The record is, moreover, verifiable without trusting its author. Any reader can recompute the SHA-256 digest of the protocol file and compare it with the value stated above; any reader can verify the OpenTimestamps proof against the Bitcoin blockchain using freely available open-source tooling; and the cantonal deposit provides an institutional record independent of the association's own infrastructure.

[17] Deposit with the Legal Services of the State Council of the Canton of Ticino (Servizi giuridici del Consiglio di Stato, Bellinzona), 28 April 2025.
[18] OpenTimestamps (opentimestamps.org) anchors the SHA-256 digest of a file in the Bitcoin blockchain, so that the existence of the exact file content at the certified date can be verified independently of the issuing party.

Each element of the record was chosen so that it can be placed before a review body, a standardisation process or an intergovernmental forum and examined on its own terms, without depending on any assertion by the issuing association.

What documented existence does change is how the state of the art is described. For as long as no candidate standard exists, the absence of surrender recognition in an autonomous system can be characterised as an inherent technical limitation: there was nothing specified to recognise, and the obligation of Article 41, though binding, had no validated, operationally available means of implementation at machine speed. Once a candidate standard is publicly available, dated and integrity-protected, that characterisation is no longer complete as a description of the facts. Whether to examine the candidate, test it, adopt it, improve it or reject it on stated grounds becomes a decision — one that can be made well or badly, but that can be documented and reviewed. The existence of a specified candidate standard shifts the assessment of non-recognition from unavoidable technical limitation to a choice subject to legal scrutiny. One clarification prevents this point from being overread: a reasoned determination, after examination, that the candidate is technically unsound would itself be the examination this argument contemplates — the record would then show a standard proposed, tested and rejected on stated grounds, and that is a legitimate outcome. What the record precludes is only one assertion: that there was nothing to examine.

Article 36 of Additional Protocol I supplies the established model for this shift. A High Contracting Party studying, developing, acquiring or adopting a new weapon, means or method of warfare "is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited" by the Protocol or by any other applicable rule.[19] The chain from that text to the present argument is short, and every link is in the treaty. Article 36 obliges the reviewing State, in the case of States Parties to the Protocol, to determine whether employment would be prohibited; Article 41 prohibits attacks against a person who should be recognised to be hors de combat. The capacity of the system under review to make such a recognition therefore falls within the subject matter of the review, rather than constituting an optional annex to it. Moreover, any determination of what is feasible must necessarily be made by reference to the state of the art, including what is publicly available at the time of the review. Nothing in this chain requires Article 36 to name third-party candidate safeguards, and this paper does not read it as doing so — the treaty text nowhere mentions them, and no such requirement is asserted. The argument is narrower: a review that addresses whether employment would violate Article 41 at machine speed has a defined question before it, and the publicly available material bearing on that question now includes a candidate means specified in testable form. This paper adds only the fact and its date: from May 2025, and in summarised form from May 2026, such a candidate has been publicly available. What follows from that fact, and in which forum, is not for this paper to say.

5. Limits

This Section states what the protocol does not resolve. The failure modes are those identified generally in Paper I, Section VII,[20] here made concrete for this specification, together with the limit that conditions everything else: adoption is at present voluntary.

False positives. A 7 Hz white light can occur incidentally — emergency strobes, damaged luminaires, reflections of periodic sources — and 144.0 MHz lies in an allocated amateur band carrying legitimate traffic. The dual-channel conjunction is designed to make coincidence rare, but how rare is an empirical question, and the datasets needed to answer it do not exist. The operational cost of a false positive is an unnecessary abstention; a protocol whose false-positive rate is high would create precisely the pressure to ignore it that a standard is intended to prevent. The number matters, and it is currently unknown.

False negatives. Emission can fail — equipment damage, power loss, terrain masking, jamming — and detection can fail even when emission succeeds. Two consequences must be reflected in the design of any recognition logic. First, the absence of the signal carries no adverse inference: Article 41 protects any person who clearly expresses an intention to surrender by whatever means, and HF SIGNAL 01

[19] AP I, Art. 36. On the reading of Art. 36 adopted here — that the review obligation makes the treatment of known, available means of compliance examinable — see Paper I, supra note 1.

[20] Paper I, supra note 1, Section VII.

adds a machine-legible means without narrowing the class of valid expressions. A system must never be configured to treat non-emission as an indicator of hostility. The protocol is a floor for legibility, not a filter on protection. Second, a detected signal that is subsequently accompanied by hostile acts falls under Article 41's own proviso, which conditions protection on abstention from hostile acts;[21] the protocol changes nothing in that respect.

Adversarial exploitation. The signal can be feigned. Existing law already addresses the gravest form of this. Article 37(1) prohibits killing, injuring or capturing an adversary by resort to perfidy, defined as "[a]cts inviting the confidence of an adversary to lead him to believe that he is entitled to, or is obliged to accord, protection under the rules of international law applicable in armed conflict, with intent to betray that confidence", and lists as its first example "the feigning of an intent to negotiate under a flag of truce or of a surrender[22]". Falsely emitting HF SIGNAL 01 in order to invite reliance and then kill, injure or capture engages these elements irrespective of the signal's status: the perfidy prohibition attaches to the feigning of surrender as such, whatever code the feigning uses. The protocol therefore creates no new protection and no new vulnerability in perfidy terms — consistently, again, with the method of this paper, which asserts no new obligation. Two clarifications are necessary. Perfidy stricto sensu requires the result element of killing, injuring or capturing; false emission solely to evade engagement does not meet that definition, although it degrades the reliability of the signal for everyone and may engage other rules on the improper use of signs and signals. And nothing in this analysis assimilates HF SIGNAL 01 to the recognised emblems of Article 38: it is not one, and its misuse is not their misuse. What adversarial pressure ultimately demands is technical: recognition logic robust to spoofing attempts, evaluated on adversarial samples — which returns the question to the datasets and validation that Section 3 records as absent.

Saturation. The operational form of the adversarial problem is broader than perfidy, and it deserves its own name. If recognition-for-abstention became widespread, an adversary would acquire an incentive not merely to feign the signal occasionally but to saturate: to field false emitters in numbers, converting every abstention into cover and every confirmed signal into noise. Deliberate saturation is the adversarial form of the false-positive problem, and it must be said plainly that no signal-based scheme is immune to it — the same tactic exists, and has always existed, against the white flag itself, and the law's response was to criminalise the abuse rather than abandon the signal. The protocol neither solves this problem nor aggravates it; what it changes is that robustness under saturation becomes a measurable property rather than a matter of speculation. The adversarial model and decoy conditions specified in Section 3 exist precisely to measure it, and the design elements that bound the exposure are already part of the specification: the dual-channel conjunction raises the cost of each false emitter above that of a stray light or a bare transmission; the good-faith activation condition identifies mass false emission as abuse on the face of the specification; and the Article 41 proviso means that abstention is revocable the moment the supposed surrender is accompanied by hostile acts. The exposure is bounded, not eliminated: whether the residual exposure is operationally tolerable is exactly the kind of question that belongs to validation and to the competent fora, not to the assertions made in this paper.

One consequence of that exposure should be stated, since it is a response that an operational command might adopt: a detector already fitted could be disabled in software, or the relevant band filtered, in order to remove the tactical liability. Nothing in this paper suggests that such a step could not be justified in a given theatre. It should, however, be observed that deliberate disabling constitutes a dated instruction attributable to an identifiable authority and is therefore more readily examinable than an absence of capability decided at the design stage and never recorded. Suppression of a recognition capability is not a return to the position that existed before the capability was introduced.

Verification of adaptive systems. Where a system's behaviour evolves in operation, its response interval is not a constant but a distribution, and no settled method exists for establishing that distribution by inspection of the system itself. This bears on the pathways unevenly. Establishing what a behavioural recogniser will do on inputs it has not seen requires reasoning about the model; establishing whether a

[21] AP I, Art. 41(2), final proviso.
[22] AP I, Art. 37(1).

system responds to a defined signature does not, because the test emits the signature and measures the response, treating the system as opaque by design. That property is not incidental — it is what allows the test to be performed by a party with no access to source, weights or design documentation, which is the ordinary condition of any external assessment of a weapon system. The limits of the present specification are unaffected: a black-box test establishes what a system did on the trials conducted, not what it will do in general.

Photosensitivity. The optical channel operates at a pulse frequency above the threshold conventionally treated as safe for photosensitive individuals: general guidance on flashing light places that threshold at three flashes per second, and 7 Hz lies above it. Photosensitive epilepsy affects a small proportion of the population, but the populations most exposed to the systems this protocol addresses are broad and unselected, and a protective signal capable of provoking a seizure in some of those it is meant to protect is a defect rather than a trade-off. No assessment of this risk has been carried out for the specification, and none is claimed here. The available mitigations are ordinary — a reduced duty cycle, a lower pulse frequency, a bounded emission duration, an intensity ceiling at the relevant distances, or a combination — and each interacts with detectability, which is why the question belongs to the validation campaign described in Section 3 rather than to an assertion in this paper. Until it is resolved, the pulse frequency stated in Section 2.1 should be read as a testable value subject to a safety constraint not yet specified.

Equipment dependence. The active electronic modality presupposes that the protected person possesses a functioning, powered emitter at the moment of encounter. The hardware is deliberately minimal, but minimal is not universal: the people most exposed to autonomous systems — displaced civilians, populations under siege, persons fleeing — are precisely those least likely to carry any equipment at all. A machine-readable standard protects only those able to speak it. This is a structural limit of any active signal-based approach. To mitigate this, the protocol's architecture includes a passive visual modality (a white disc on a red field) that requires no power, equipment, or emission (see the accompanying Architecture Note). While this passive modality addresses the equipment-dependence objection for those who have nothing, its machine-perception parameters (chromaticity, minimum size, viewing angle) remain to be specified and validated. The active electronic modality described in this paper remains subject to the equipment-dependence limit stated above. In either case, the protocol can only ever be a floor for legibility, never a filter on protection, and no inference of any kind may be drawn from its absence.

Geometry of the passive modality. The passive modality is specified as a white disc on a red field. That choice is a placeholder: it was not derived from a stated perceptual or legal constraint, and this paper advances no argument for it. Two considerations bear on its eventual determination. First, machine distinctiveness: a filled light disc on a uniform darker field is a common configuration in the ordinary visual world — signage, logos, apertures — and the false-positive exposure of the passive modality has not been characterised. Second, and independently, the pairing of white and red in a centred figure-on-field arrangement is the chromatic signature of the distinctive emblems protected under Article 38 of Additional Protocol I. This paper asserts no assimilation to those emblems and claims none of their protection; but a protective sign intended for use by untrained persons, without instruction, should be distinguishable from them at the distances and under the conditions in which both are actually read. Whether the eventual geometry retains the present colours is a question for a standardisation process, in which distinctiveness from the Article 38 emblems belongs among the criteria alongside chromaticity, minimum size and viewing angle.

Voluntariness of adoption. No instrument in force today obliges any manufacturer or any State to implement recognition of HF SIGNAL 01, or of any other specific code. Adoption is voluntary, and this paper does not claim otherwise — it could not honestly do so. Its claims end where they began: the underlying obligations exist, and long predate the systems at issue; a candidate means of giving effect to them at machine speed now exists in documented, testable form; and whether the gap between the two is closed — by this specification, by a better one, or by a standard developed in the competent institutions — is a question this paper can open but not decide.

6. Conclusion: An Invitation, Not a Product

This paper has deliberately made limited claims. It has not claimed that HF SIGNAL 01 works: no measurement exists. It has not claimed that anyone must adopt it: no instrument requires that. It has claimed that an obligation long in force — to recognise the person who clearly expresses an intention to surrender, and to do everything feasible to verify — lacks, in encounters decided at machine speed, a validated and operationally available means of implementation; that a candidate means now exists in a publicly deposited, integrity-protected form, with its active modality specified at the level of parameters that can be tested; and that, from the moment such a candidate exists, the absence of surrender recognition is no longer only a fact about technology, but also a fact about decisions.

A final clarification is owed to readers who locate the problem elsewhere — who hold that the real issue is autonomy in targeting as such, and that the answer is stronger human control or prohibition rather than new signals. Nothing in this paper is an argument for autonomous engagement, and nothing in it should be quoted as one. The protocol is not a licence: recognition capability is compatible with, and does not substitute for, requirements of human control, and Section 1 lists the restoration of human decision time as the first of the paths — the one that dissolves the problem this paper addresses. If such systems were prohibited, or confined under supervision that makes the human decision real, the urgency of a machine-readable standard would recede accordingly, and the author would not mourn it. This paper takes no position on that policy question. It addresses the world in which such systems exist and are being deployed — a world in which the absence of any means of expressing surrender to them protects no one, and constrains no one. And to readers who would prefer the argument for a standard without the candidate itself — a methodology for developing a machine-readable surrender standard, rather than a completed proposal — the answer is that a standards process needs an object to work on. Methodologies without artefacts produce committees; artefacts submitted to processes are how standards have historically begun, the distinctive signals of Annex I among them. This paper supplies the object and, in the same act, submits it.

The natural next steps do not belong to the author. They are, first, independent technical validation by a university laboratory in machine perception or robotics, or by a standardisation body, against the criteria set out in Section 3, beginning with an explicit adversarial model, dataset construction, and latency measurement; and, second, examination in the competent fora, in particular the Group of Governmental Experts on Lethal Autonomous Weapons Systems, where the question of machine-readable expression of surrender belongs on the agenda whether or not this specification survives scrutiny. The specification is offered for testing, amendment, or supersession. What should not survive its publication is the claim that there was nothing to recognise.

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